Route Wise — Market Requirements Document (MRD)

DOCUMENT: MRD-v2.0
TEAM: Cyber Knight
AUDIENCE: Mentors & Judges
STATUS: Final Pitch Ready

The Business Case, Market Need & Real-World Problem — Explaining Route Wise in Simple English

The 30-Second Elevator Pitch for Judges: Most navigation apps blindly suggest the shortest route, even if it leads straight into heavy junction gridlock. **Route Wise (Team Cyber Knight)** is an intelligent traffic advisory platform and live dashboard that evaluates multiple driving alternatives, penalizes severe junction bottlenecks, and predicts 15-minute congestion trends using an official public **ARIMA (1,0,1)** ML model from GitHub. It explains *why* it chose the route in plain words (e.g. *'Saves 14 mins travel time and 37 km shorter vs Route #2'*), saving time, fuel, and driver frustration for daily commuters and couriers.

1. Real-World Stories: Why Traditional GPS Fails Commuters & Couriers

Story 1: Karthik (Daily Commuter, 9:00 AM Office Cutoff): Trapped at 5:15 PM peak rush from Central Bus Stand to Cantonment. Traditional GPS recommends the direct arterial (10.0 km) with unexpected junction bottleneck (+18m delay, 42m total). Route Wise evaluates 3 alternatives and diverts him via the outer ring corridor (12.5 km, +2m delay, 26m total), saving **16 minutes**.
Story 2: Selvi (Quick-Commerce Delivery Rider, 15-Min SLA): Carrying 3 grocery orders at 8:30 PM. Traditional GPS sends her through Main Bazaar road (3.2 km), but a 3-cycle junction signal adds +9m delay (17m total, breaching delivery SLA). Route Wise detects the bottleneck and reroutes Selvi via a 4.1 km parallel residential bypass corridor (8m total, 0m delay). She arrives with **7 minutes to spare**, earning on-time incentives!

Persona & Journey	GPS Strategy	Distance	Delay	Total Time	Outcome & Business Impact
Karthik (Office Commute)	Standard GPS (Shortest)	10.0 km	**+18m**	**42 mins**	Late arrival, high stress, fuel wasted idling in junction.
Karthik (Office Commute)	**Route Wise AI (ARIMA + Scoring)**	12.5 km	**+2m**	**26 mins**	**16m faster**, on-time punch-in, constant fuel-efficient speed.
Selvi (Quick-Commerce)	Standard GPS (Shortest)	3.2 km	**+9m**	**17 mins**	**SLA Breached**, customer rating penalty, lost incentive.
Selvi (Quick-Commerce)	**Route Wise AI (Bypass Corridor)**	4.1 km	**+0m**	**8 mins**	**SLA Achieved (7m early)**, zero junction stops, peak bonus earned.

2. The Big Market Problem We Solve

1. Opaque 'Black-Box' Algorithms: Traditional apps give a single route without explaining why it was picked, leaving commuters uncertain.
2. Inability to Predict Traffic Spikes: Existing maps show current delay, but cannot tell if congestion will clear or worsen in 15 minutes.
3. Demo & Network Fragility: Standard mapping APIs crash when offline or over quota. Developers need high-reliability fail-safes.

3. Market Size: Multi-Billion Dollar Opportunity

TAM: $14.2 Billion Global GIS, Navigation & Fleet Logistics API Market	**SAM: $2.8 Billion** Urban Last-Mile Delivery & Commuter Routing Software	**SOM: $350 Million** Tier-2 & Tier-1 Indian Logistics SaaS, Startups & Commuters

4. Target Audience, Competition & Revenue Model

4.1 Who Needs This? (Target Customer Segments)

- **Daily Urban Commuters (2W & 4W):** Need to bypass known junction bottlenecks during 8:00 AM – 10:00 AM and 5:00 PM – 7:30 PM peak commutes.
- **On-Demand Couriers & Delivery Drivers:** Under strict SLAs (8 deliveries/hr). A 15-minute junction delay causes missed order deadlines and penalties.
- **Fleet & Logistics Dispatchers:** Managing delivery vans. A 15-minute delay per driver compounds into thousands of wasted fuel rupees weekly.

4.2 Why Route Wise Wins Against Big Tech APIs

Feature / Capability	Google Maps API	Mapbox Navigation	Route Wise (Cyber Knight)
Configurable Scoring Weights	No (Fixed default algorithm)	Partial (Limited vehicle profiles)	**Yes — Customize Time, Delay Penalty & Distance**
Transparent 'Why' Rationale	No (Opaque black-box route)	No (No explanation given)	**Yes — Clear tags like 'Saves 14m & 37 km shorter'**
15-Min ML Traffic Forecast	No (Live conditions only)	No (Historical averages only)	**Yes — Public ARIMA(1,0,1) model for +15m, +30m, +45m**
Traffic Severity Badges	Partial (Visual map lines only)	Partial (Generic speed colors)	**Yes — Green / Yellow / Red delay classifier**
Offline Demo Fail-Safe	No (Fails if internet drops)	No (Requires paid offline packs)	**Yes — High-fidelity local coordinate dictionary**
Side-by-Side Comparison Matrix	No (Must switch back & forth)	No (Single route view)	**Yes — Simultaneous comparison table**

4.3 How We Monetize (Simple Tiered Model)

Tier	Price	Included Quota & Capabilities	Target Customer
Free Developer	$0 / mo	1,000 queries/month, 60 req/min rate limit, full dashboard access, offline mock mode.	Hackathons, student projects, prototyping.
Growth Fleet	$0.003 / query	Up to 250,000 queries/month, custom scoring presets, 60-second live refresh API, ARIMA forecast.	Courier startups, local delivery fleets.
Enterprise	Custom SLA	Unlimited queries, multi-stop TSP route optimization, dedicated container hosting, 99.9% uptime SLA.	Logistics corporations, mobility providers.

4.4 Top 3 Takeaways to Explain to Judges & Mentors

1. Not Just Another Map: We don't reinvent map tiles; we solve the intelligence layer that evaluates multiple alternatives and tells you *why* Route #1 is the best.	**2. Tangible Commuter ROI:** Saving 15 minutes of junction delay per trip directly reduces fuel consumption, vehicle wear, late penalties, and driver burnout.	**3. Production-Ready Tech:** Powered by TomTom live traffic, public **ARIMA (1,0,1)** ML forecasting, resilient fail-safes, and **40 automated tests** passed at 100%.